A five-colour lower bound for the regular fourteen-gon norm

Bartosz Paprzycki
Independent

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Abstract

Let P be a regular fourteen-gon centred at the origin. We prove that every colouring of the plane in four colours has a monochromatic pair at distance one in the norm whose closed unit ball is P. The proof uses a graph on 1 540 labelled vertices with 13 755 edges, realised at unit distance by coordinates in $\mathbb{Z}[e^{\pi i/7}]$. Exact algebraic witnesses certify every edge, and a propositional refutation rules out a proper four-colouring of the graph. The geometric argument, finite obstruction and transfer to the plane are formalised in Lean. Together with Gehér's upper bound, the result gives $5 \le \chi(\mathbb{R}^2, P) \le 6$.

1 Introduction

For a centrally symmetric convex body $P \subset \mathbb{R}^2$ with the origin in its interior, let $\chi(\mathbb{R}^2, P)$ denote the least number of colours needed to colour the plane so that points at distance one in the norm with unit ball P receive different colours. No measurability condition is imposed on the colour classes.

Exoo, Fisher and Ismailescu proved a lower bound of five when the unit ball is a regular octagon, decagon or dodecagon [1]. Their Problem 5.1 asks whether the same lower bound holds for every regular $2n$-gon with $n \ge 4$, and more generally for centrally symmetric polygons with that many sides. We establish the regular fourteen-gon case by constructing a finite unit-distance graph with no proper four-colouring. The construction follows their sumset strategy; the geometry and colouring obstruction are verified in Lean.

Identify $\mathbb{R}^2$ with $\mathbb{C}$ and set

$$\zeta = e^{\pi i/7}, \qquad P = \operatorname{conv}_{\mathbb{R}}\{1, \zeta, \dots, \zeta^{13}\}, \qquad \|z\|_P = \inf\{r > 0 : z \in rP\}.$$

The set P is a regular fourteen-gon of circumradius one. Changing its scale or orientation gives an isometric normed plane after the corresponding linear change of coordinates.

Theorem 1. *For every map $c : \mathbb{C} \to \{0, 1, 2, 3\}$, there are $x, y \in \mathbb{C}$ such that $\|x - y\|_P = 1$ and $c(x) = c(y)$. In particular, $\chi(\mathbb{R}^2, P) \ge 5$.*

Gehér proved that six colours suffice for every regular polygon with an even number of vertices at most 22 [2, Theorem 1]. Thus:

Corollary 2. $5 \le \chi(\mathbb{R}^2, P) \le 6$.

We do not determine which of these two values is correct.

The proof combines an exact unit-distance realisation (Sections 2–3) with a machine-checkable refutation of the graph's four-colouring formula (Section 4). A colouring of the plane would restrict to a colouring of this graph; Section 5 gives the resulting contradiction.

2 The polygon norm and its unit sides

The set P is compact, convex and symmetric under $z \mapsto -z$, since $\zeta^7 = -1$. The vectors 1 and ζ are linearly independent over $\mathbb{R}$, so $\operatorname{conv}\{1, -1, \zeta, -\zeta\} \subset P$ contains a neighbourhood of zero. Hence the gauge $\|\cdot\|_P$ is finite. Convexity gives the triangle inequality, symmetry gives absolute homogeneity, and boundedness gives positive definiteness. Thus this gauge is a norm. Compactness also gives $\{z : \|z\|_P \le 1\} = P$.

The following elementary fact is the geometric part of the edge checker.

Lemma 3. *For $j \in \mathbb{Z}$ and $0 \le t \le 1$,*

$$\left\|\zeta^j\big((1-t)+t\zeta\big)\right\|_P = 1.$$

Proof. Multiplication by ζ permutes the vertices of P, hence preserves P and its gauge. It is enough to treat $j = 0$. Put $\theta = \pi/7$ and define the real linear functional

$$L(z) = \operatorname{Re} z + \operatorname{Re}(\overline{\zeta} z), \qquad h = 1 + \cos\theta > 0.$$

For a vertex ζ^k,

$$L(\zeta^k) = \cos(k\theta) + \cos((k-1)\theta) = 2\cos(\theta/2)\cos((k-\tfrac{1}{2})\theta) \le h.$$

The last inequality holds because the angles $(k - \frac{1}{2})\theta$, modulo 2π, are at least $\theta/2$ from zero. Equality holds for $k = 0, 1$. By convexity, $L(z) \le h$ for every $z \in P$.

Now $s = (1-t) + t\zeta$ belongs to P and satisfies $L(s) = h$. Membership gives $\|s\|_P \le 1$. If $s = ry$ with $r > 0$ and $y \in P$, then $h = rL(y) \le rh$, so $r \ge 1$. Taking the infimum over all such r proves $\|s\|_P \ge 1$. □

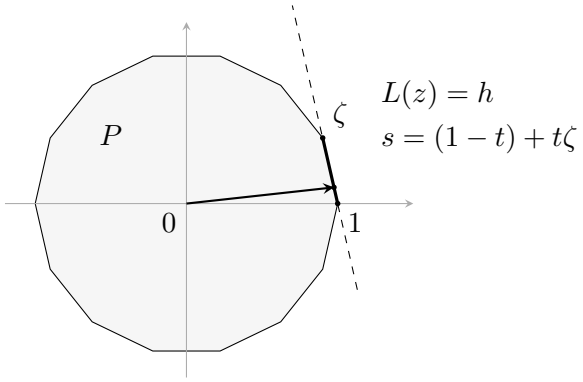


Figure 1: The side from 1 to ζ lies on a supporting line of P. Every point on this side has polygon norm one, although its Euclidean distance from zero need not be one.

3 Exact coordinates for the finite graph

3.1 Arithmetic in the cyclotomic coordinates

For $a = (a_0, \dots, a_5) \in \mathbb{Z}^6$, write $\operatorname{ev}(a) = \sum_{k=0}^{5} a_k \zeta^k$. The identity

$$\zeta^6 - \zeta^5 + \zeta^4 - \zeta^3 + \zeta^2 - \zeta + 1 = 0 \tag{1}$$

follows from $\zeta^7 = -1$ and $\zeta \ne -1$. Consequently multiplication by ζ can be performed on integer tuples by the map

$$R(a) = (-a_5,\ a_0 + a_5,\ a_1 - a_5,\ a_2 + a_5,\ a_3 - a_5,\ a_4 + a_5). \tag{2}$$

In particular $\mathrm{ev}(R(a)) = \zeta\,\mathrm{ev}(a)$. Subtraction of tuples also commutes with evaluation.

The side parameters belong to a real cubic field. Set

$$\rho = 2\cos(2\pi/7) = \zeta^2 - \zeta^5.$$

The second equality views the real number ρ as a complex number. Reduction using (1) gives

$$f(\rho) = 0, \qquad f(X) = X^3 + X^2 - 2X - 1. \tag{3}$$

Moreover $\rho > 1$, because $2\pi/7 < \pi/3$. The polynomial f is strictly increasing on $[1, \infty)$: $f'(X) = 3X^2 + 2X - 2 > 0$ there. Exact substitution into f yields $f(\ell) < 0 < f(u)$ for

$$\ell = \frac{1246979603717467}{10^{15}}, \qquad u = \frac{1246979603717468}{10^{15}}. \tag{4}$$

It follows that $\ell < \rho < u$. This argument identifies the intended root of (3) from its geometric definition.

For integers A, B, C, let $t = A + B\rho + C\rho^2$. A tuple representing the point $(1 - t) + t\zeta$ is

$$S(A, B, C) = (1 + B - A - 2C,\ A + 2C - B,\ 0,\ C,\ B - 2C,\ C). \tag{5}$$

Indeed, substituting $\rho = \zeta^2 - \zeta^5$ and reducing by (1) gives $\mathrm{ev}(S(A, B, C)) = (1 - t) + t\zeta$.

3.2 The coordinate and witness lists

Let $V = \{0, \ldots, 1539\}$. The coordinate list in the accompanying file `minkowski14_final_graph.json` assigns a tuple $a_v \in \mathbb{Z}^6$ to each $v \in V$; put $p_v = \mathrm{ev}(a_v)$. Its edge list defines a graph $G = (V, E)$ with $|E| = 13\,755$. Vertex labels are zero based. The files are described in Appendix A, and the complete formal sources are available in the repository [4].

For each listed edge (v, w) there is a witness (j, r), where $0 \le j < 14$ and r indexes one of the 29 triples in Table 1. The witness asserts the integer equality

$$a_v - a_w = R^j(S(A_r, B_r, C_r)). \tag{6}$$

Thus verification of an edge reduces to six integer equalities and an interval bound for $t_r = A_r + B_r\rho + C_r\rho^2$.

r	A_r	B_r	C_r	r	A_r	B_r	C_r	r	A_r	B_r	C_r
0	-4	0	3	10	-1	1	0	20	2	-1	0
1	-4	1	2	11	0	-2	2	21	2	0	-1
2	-3	-1	3	12	0	-1	1	22	2	1	-2
3	-3	0	2	13	0	2	-1	23	3	-1	-1
4	-2	-3	4	14	0	3	-2	24	3	2	-3
5	-2	-2	3	15	1	-3	2	25	3	3	-4
6	-2	1	1	16	1	-2	1	26	4	0	-2
7	-2	2	0	17	1	0	0	27	4	1	-3
8	-1	-1	2	18	1	1	-1	28	4	2	-4
9	-1	0	1	19	1	2	-2				

Table 1: The triples used to certify the edges, with the same indices as in the formal development. Each gives $0 \le t_r \le 1$.

Lemma 4. *For each triple in Table 1,* $0 \le A_r + B_r\rho + C_r\rho^2 \le 1$.

Proof. Since $0 < \ell < \rho < u$, a lower and upper bound for any such expression are

$$L_r = A_r + \min(B_r\ell, B_r u) + \min(C_r\ell^2, C_r u^2),$$
$$U_r = A_r + \max(B_r\ell, B_r u) + \max(C_r\ell^2, C_r u^2).$$

Substitution of the 29 triples gives $L_r \ge 0$ and $U_r \le 1$. These are rational inequalities with the fixed denominator 10^{30} after clearing fractions. The formal theorem `parameter_bounds` checks all cases using (4). □

Proposition 5 (Certified edge list)**.** *Every* $(v, w) \in E$ *satisfies* $\|p_v - p_w\|_P = 1$.

Proof. The certificate check verifies that every $(v, w) \in E$ has a witness (j, r) satisfying (6). This is a finite check of six integer equalities per edge. Evaluating these equalities gives

$$p_v - p_w = \zeta^j((1 - t_r) + t_r\zeta).$$

Apply Lemmas 4 and 3. □

For example, the edge $(0, 106)$ has

$$a_0 = (-6, 4, 0, 1, -4, 1), \qquad a_{106} = (-3, 2, 0, 1, -3, 1).$$

Its witness is $j = 7$, $r = 10$, so $t_r = \rho - 1$ and $S(-1, 1, 0) = (3, -2, 0, 0, 1, 0)$. Since $\zeta^7 = -1$, equality (6) gives $p_0 - p_{106} = -((2 - \rho) + (\rho - 1)\zeta)$. The interval (4) puts $\rho - 1$ strictly between zero and one, so this edge corresponds to a point in the interior of a polygon side, rather than to a polygon vertex.

3.3 How the witness was obtained

Following the sumset strategy of Exoo, Fisher and Ismailescu [1], the search used the side parameters

$$T = \{0, \rho - 1, \rho^2 - \rho, \rho^2 - 1, 1 + \rho - \rho^2\}.$$

Each parameter lies in $\mathbb{Z}[\rho] \cap [0, 1]$, as follows from (4). Thus the generating set

$$U = \{\zeta^j((1 - t) + t\zeta) : t \in T,\ 0 \le j < 14\}$$

consists of unit vectors with exact integer-tuple coordinates given by (5) and (2). Forming $U + U$ and testing differences can therefore be done in the same exact arithmetic. The search retained unit-distance edges and used unsatisfiable SAT cores to reduce the graph to the supplied coordinate and edge lists.

4 The four-colouring obstruction

For $v \in V$ and $c \in \{0, 1, 2, 3\}$, introduce a Boolean variable $x_{v,c}$ meaning that vertex v has colour c. Let F_G be the conjunction of the following clauses:

$$x_{v,0} \vee x_{v,1} \vee x_{v,2} \vee x_{v,3} \qquad (v \in V), \tag{7}$$
$$\neg x_{v,c} \vee \neg x_{v,d} \qquad (v \in V,\ 0 \le c < d \le 3), \tag{8}$$
$$\neg x_{v,c} \vee \neg x_{w,c} \qquad ((v, w) \in E,\ 0 \le c \le 3), \tag{9}$$
$$x_{759,0}. \tag{10}$$

There are $4|V| = 6\,160$ variables and

$$|V| + 6|V| + 4|E| + 1 = 1\,540 + 9\,240 + 55\,020 + 1 = 65\,801$$

clauses. In DIMACS notation, $x_{v,c}$ has index $4v + c + 1$; the last clause is therefore `3037 0`.

Lemma 6. *If G has a proper four-colouring, then F_G is satisfiable.*

Proof. Permute the four colour names so that vertex 759 has colour zero. Assign $x_{v,c}$ to be true precisely for the assigned colour of v. Clauses (7) and (8) express that each vertex has exactly one colour, clauses (9) hold by properness, and (10) holds by the permutation. □

Proposition 7 (Certified refutation)**.** *The formula F_G is unsatisfiable. Consequently G has no proper four-colouring.*

Computer-assisted proof. The file `minkowski14_final_4color.lrat.gz` contains an LRAT refutation of the exact formula in `minkowski14_final_4color.cnf`. LRAT is a clausal proof format that supplies hints for checking each added clause [3]. The trace is imported in 42 parts as ordinary Lean proof expressions; the final part derives the empty clause. The formal encoding theorems identify the imported formula with the clauses (7)–(10), including the anchor condition. Lean checks both this identification and the refutation. Hence F_G is unsatisfiable, and Lemma 6 rules out a proper four-colouring of G. □

The supplied trace allows this derivation to be checked independently of the search and the SAT solver.

5 Transfer to the plane

Proof of Theorem 1. Suppose that $c : \mathbb{C} \to \{0, 1, 2, 3\}$ assigns different colours to every pair at polygon distance one. Colour each vertex v of G by $c(p_v)$. Proposition 5 makes this a proper four-colouring of G, contradicting Proposition 7. □

The listed edges suffice for the contradiction, even if other pairs among the coordinates also have unit distance. The pullback colouring is well defined even when two vertex labels represent the same point; adjacent vertices have distinct images because their distance is one. A colouring with fewer than four colours also gives a map to $\{0, 1, 2, 3\}$, so the argument rules out all such colourings.

6 Formalisation and reproducibility

The formalisation uses Lean 4.34.0 and mathlib at commit

`5ed2965256430c3649e86755f9576b54eca72435`.

The type `Minkowski14.Plane` is the complex plane equipped with the polygon norm. Theorems `plane_dimension`, `plane_unit_ball` and `plane_distance` establish, respectively, its real dimension two, its closed unit ball P, and its distance function $d(x, y) = \|x - y\|_P$. These theorems identify the formal normed space with the geometric plane used in Theorem 1.

The final theorem, `Minkowski14.at_least_five_colours`, states that for every natural number n and every map c from this plane to `Fin n`, the hypothesis

$$\forall x, y, \quad d(x, y) = 1 \implies c(x) \neq c(y)$$

implies $5 \leq n$. Here `Fin n` is the type of natural numbers strictly less than n.

A public build on 6 October 2026 compiled all 162 project modules from source and rejected three deliberately invalid controls: a false coordinate equality, a false SAT refutation, and a proof expression of the wrong type. A valid small SAT control passed. The source files are matched to the compiler records by SHA-256 hashes.

The final theorem depends only on the standard Lean axioms `propext`, `Classical.choice` and `Quot.sound`. The project proof has no additional axioms, admissions, `unsafe` declarations

Mathematical component	Formal sources under `formal/`
Polygon norm and supporting side	`Polygon.lean`, `Facet.lean`
Cyclotomic arithmetic and parameter bounds	`Algebra.lean`, `Coordinates.lean`, `Parameters.lean`
Coordinate and edge certificates	`Positions.lean`, `Geometry/`, `Embedding.lean`
Colouring formula and refutation	`MathIsEasy/Minkowski14/`, `MathIsEasy/Minkowski14LRAT/`
Normed plane and final transfer	`Plane.lean`, `ColouringTransfer.lean`, `Final.lean`

Table 2: Where to compare the mathematical statements with their formal versions.

or uses of `native_decide`. Proof expressions are checked by Lean's ordinary kernel. This verification relies on the kernel and the correspondence between the formal and mathematical statements. No separate kernel implementation was used.

The repository includes the pinned dependencies, build procedure, compiler records, axiom output and failed-control diagnostics [4]. Its lightweight integrity command checks hashes and correspondence of the published files without compiling Lean. Rebuilding the proof is a separate operation, described in `REPRODUCE.rst`.

7 Further questions

Does the entire plane admit a proper five-colouring? How small can a finite unit-distance obstruction to four colours be? Extending the construction to other regular polygons would address further cases of [1, Problem 5.1]; the coordinates and side parameters used here are specific to the fourteen-gon.

Use of AI

The author initiated and directed this work but has no background in mathematical research and did not supply the mathematical argument. The graph search, certificate generation, geometric proof, Lean formalisation and manuscript were developed with AI assistance. OpenAI Codex (GPT-6 Astra) was used for the complete geometric formalisation, verification tools and exposition. Independent human peer review of the manuscript is not claimed.

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A Data files and certificate conventions

The data files are included in the repository [4]. The graph, colouring formula and LRAT certificate are under `formal/artifacts/`; the edge witnesses are in `docs/graph.json`.

`minkowski14_final_graph.json`

The exact coordinate and edge lists. The field `vertices_power_basis` contains the six integers $(a_0, \ldots, a_5)$ for each vertex, in label order. The field `edges` contains pairs of zero-based vertex labels. Only these two fields define the graph used in the proof.

`docs/graph.json`
: The same coordinates, the ordered parameter triples, and triples (v, w, k) for the edges. Decode a witness by $r = \lfloor k/14 \rfloor$ and $j = k \bmod 14$, then check (6). Edge order may differ from `minkowski14_final_graph.json`; the edge sets agree.

`minkowski14_final_4color.cnf`
: The DIMACS formula described in Section 4.

`minkowski14_final_4color.lrat.gz`
: The gzip-compressed LRAT refutation; decompress it before replay. Its filename in the Lean project is `minkowski14_final_4color.lrat`.

`formal/`
: The complete Lean sources and pinned dependencies. The repository's `REPRODUCE.rst` gives the build instructions; precompiled project results are not required.

`evidence/`
: Source and certificate hashes, compiler records, axiom output and control results.

For comparison with the recorded build, the SHA-256 hash of the *decompressed* LRAT file is

`9ff6788806bedd8596fb8972fa546d86b6e193dc950e2542a2dcc4e29b27b2d6`

References

[1] G. Exoo, D. Fisher and D. Ismailescu, *The chromatic number of the Minkowski plane—the regular polygon case*, 2021. arXiv:2108.12861.

[2] P. Gehér, *Note on the chromatic number of Minkowski planes: the regular polygon case*, 2023. arXiv:2301.13695.

[3] L. Cruz-Filipe, M. J. H. Heule, W. A. Hunt Jr., M. Kaufmann and P. Schneider-Kamp, *Efficient certified RAT verification*, 2016. arXiv:1612.02353.

[4] B. Paprzycki (MathIsEvenEasier), *Minkowski fourteen-gon: graph, certificates and Lean formalisation*, 2026. `https://github.com/MathIsEvenEasier/minkowski14`. The recorded public source build uses commit `e0f5e2a086fb0ab080b4a9393b72d33fc87e5918`; build and compiler records.